

LINEAR COMBINATIONS OF RANDOM VARIABLES

If X and Y are independent random variables with expected values of $E(X)$ and $E(Y)$ and variances of $\text{Var}(X)$ and $\text{Var}(Y)$ respectively, then the linear combinations of the variables X and Y have expected values and variances as follows (where a , b and c are constants):

mean	
Expected value	Variance
$E(ax) = a \cdot E(x)$ ↓ dice	$\text{Var}(ax) = a^2 \cdot \text{Var}(X)$
$E(x+y) = E(x) + E(y)$ ↓ ↓ dice coin	$\text{Var}(x+y) = \text{Var}(x) + \text{Var}(y)$
$E(ax+by) = a \cdot E(x) + b \cdot E(y)$	$\text{Var}(ax+by) = a^2 \cdot \text{Var}(x) + b^2 \cdot \text{Var}(y)$
$E(x-y) = E(x) - E(y)$	$\text{Var}(x-y) = \text{Var}(x) + \text{Var}(y)$ (always add variances)
$E(x+c) = E(x) + c$ ↓ eg 15	$\text{Var}(x+c) = \text{Var}(x)$ 1 2 3 4 5 6 7 8 9 10 11

⇒ Single dice (X)

x	1	2	3	4	5	6
P	0.17	0.17	0.17	0.17	0.17	0.17

$$E(X) = [(1 \times 0.17) + (2 \times 0.17) + (3 \times 0.17) + (4 \times 0.17) + (5 \times 0.17) + (6 \times 0.17)] = 3.50$$

$$\text{Var}(X) = [(1^2 \times 0.17) + (2^2 \times 0.17) + (3^2 \times 0.17) + (4^2 \times 0.17) + (5^2 \times 0.17) + (6^2 \times 0.17)] - (3.5)^2 = 2.92$$

⇒ let a new variable = $2 \times X$

new variable	2	4	6	8	10	12
P	0.17	0.17	0.17	0.17	0.17	0.17

$$E(2X) = [(2 \times 0.17) + (4 \times 0.17) + (6 \times 0.17) + (8 \times 0.17) + (10 \times 0.17) + (12 \times 0.17)] = 7 \quad (2 \times E(X) = 2 \times 3.5 = 7)$$

$$\text{Var}(2X) = [(2^2 \times 0.17) + (4^2 \times 0.17) + (6^2 \times 0.17) + (8^2 \times 0.17) + (10^2 \times 0.17) + (12^2 \times 0.17)] - (7)^2 = 11.68 \quad (\text{Var}(2X) = 2^2 \times \text{Var}(X) = 4 \times 2.92 = 11.68)$$

COIN

Y	0	1
P	0.5	0.5

$$E(Y) = (0 \times 0.5) + (1 \times 0.5) = 0.5$$

$$\text{Var}(Y) = [(0^2 \times 0.5) + (1^2 \times 0.5)] - (0.5)^2 = 0.25$$

Dice + coin

		outcome of dice						
		1	2	3	4	5	6	
outcome of coin	0	1	2	3	4	5	6	
	1	2	3	4	5	6	7	
$x+y$								
t		1	2	3	4	5	6	7
P		$\frac{1}{12}$	$\frac{2}{12}$	$\frac{2}{12}$	$\frac{2}{12}$	$\frac{2}{12}$	$\frac{2}{12}$	$\frac{1}{12}$

$$E(t) = 4 \quad (E(x) + E(y) = 3.5 + 0.5 = 4)$$

$$\text{Var}(t) = 3.17 \quad (\text{Var}(x+y) = 2.92 + 0.25 = 3.17)$$

Example:

1. A random variable X has the following probability distribution.

outcome	1	2	3
Probability	0.4	0.4	0.2

(i) calculate $E(4X+5)$

$$E(4X+5) = (9 \times 0.4) + (13 \times 0.4) + (17 \times 0.2)$$

$$= 12.2$$

X	1	2	3
$4X+5$	9	13	17
$P(X=x)$	0.4	0.4	0.2

(ii) calculate $4E(X) + 5$

$$E(X) = (1 \times 0.4) + (2 \times 0.4) + (3 \times 0.2) = 1.8$$

$$4E(X) + 5 = (4 \times 1.8) + 5 = 12.2$$

2. The random variable X has the following probability distribution

x	1	2	3	4
$P(X=x)$	0.6	0.2	0.1	0.1

(i) Find $\text{Var}(X)$.

$$E(X) = (1 \times 0.6) + (2 \times 0.2) + (3 \times 0.1) + (4 \times 0.1) = 1.7$$

$$\text{Var}(X) = [(1^2 \times 0.6) + (2^2 \times 0.2) + (3^2 \times 0.1) + (4^2 \times 0.1)] - (1.7)^2 = 1.01$$

(ii) Find $\text{Var}(3X + 7)$

$$\text{Var}(3X + 7) = 3^2 \times \text{Var}(X)$$

$$\uparrow = 9 \times 1.01 = 9.09$$

7 doesn't affect

25.3.20

if x is normal
combination
also normal

2, 3, 4, 5, 6, 7, 8, 9, 10

THE COMBINATION: $X + X$

Note that with random variables, $X + X$ is not the same as $2X$.

Eg. If X is the outcome on a dice, then $X + X$ means throw the dice two times and add the two outcomes (giving 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 or 12) whereas $2X$ means throw dice once and multiply that outcome by 2 ($\times 2$) (giving 2, 4, 6, 8, 10 or 12)

Expected Value

Variance

$$E(X + X) = E(X) + E(X) = 2 \cdot E(X)$$

$$\text{Var}(X + X) = \text{Var}(X) + \text{Var}(X) = 2 \cdot \text{Var}(X)$$

remember that $\text{Var}(2X) = 2^2 \text{Var}(X)$

DICE ONE (X)

X	1	2	3	4	5	6
P	0.17	0.17	0.17	0.17	0.17	0.17

$$E(X) = 3.5$$

$$\text{Var}(X) = 2.92$$

DICE TWO (X)

X	1	2	3	4	5	6
P	0.17	0.17	0.17	0.17	0.17	0.17

$$E(X) = 3.5$$

$$\text{Var}(X) = 2.92$$

Let a new variable = outcome from dice 1 added to outcome from dice 2

X + X	X + X	2	3	4	5	6	7	8	9	10	11	12
P		0.03	0.06	0.08	0.11	0.14	0.17	0.14	0.11	0.08	0.06	0.03
		$\downarrow$	$\downarrow$									
		1/36	2/36									

$$E(X+X) = (2 \times 0.03) + (3 \times 0.06) + (4 \times 0.08) + \dots + (12 \times 0.03)$$

$$= 7.00 = 2 \times E(X) = 2 \times 3.5$$

$$\text{Var}(X+X) = [(2^2 \times 0.03) + (3^2 \times 0.06) + (4^2 \times 0.08) + \dots + (12^2 \times 0.03)] - (7)^2$$

$$= 5.83 = 2 \times \text{Var}(X) = 2 \times 2.92$$

LINEAR COMBINATIONS OF NORMALLY DISTRIBUTED RVs

If the variable X and Y are independent and normally distributed, then the distributions of $(X+Y)$ and $(X-Y)$ are also normal.

The means of these distributions are $E(X) + E(Y)$ and $E(X) - E(Y)$

You must, however, be careful when you come to their variances, since $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$

Example 1

Robert Fisher, a keen chess player, visits his local club most days. The total time taken to drive to the club and back is modelled by a normal variable with mean 25 minutes and standard deviation 3 minutes. The time spent at the chess club is also modelled by a normal variable with mean 120 minutes and standard deviation 10 minutes. Find the prob. that on a certain evening Mr. Fisher is away from home for more than $2\frac{1}{2}$ hours.

Solution:

Let the random variable $X \sim N(25, 3^2)$ - driving time.

$Y \sim N(120, 10^2)$ - time spent at the chess club.

Then the random variable T , where $T = X + Y \sim N(145, 109)$ ^{total time}

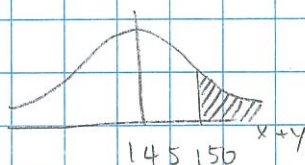
$$E(X+Y) = 25 + 120 = 145$$

$$\text{Var}(X+Y) = 3^2 + 10^2 = 109$$

So, the prob. that Mr. Fisher is away for more than $2\frac{1}{2}$ hours is

$$2\frac{1}{2} \text{ hours} = 150 \text{ mins.}$$

$$\begin{aligned} P(X > 150) &= P\left(Z > \frac{150 - 145}{\sqrt{109}}\right) \\ &= P(Z > 0.479) = 0.316 \end{aligned}$$



Ex 2.

In the manufacture of a bridge made entirely from wood, circular pegs have to fit into circular holes. The diameters of the pegs are normally distributed with mean 1.60 cm and standard deviation 0.01 cm, while the diameters of the holes are normally distributed with mean 1.65 cm and SD 0.02 cm. What is the probability that a randomly chosen peg will not fit into a randomly chosen hole?

Solution

Let $X \sim N(1.65, 0.02^2) = N(1.65, 0.0004)$ diameter of hole
 $Y \sim N(1.60, 0.01^2) = N(1.60, 0.0001)$ diameter of peg

Let $F = X - Y$ - the gap remaining between the peg and the hole
So the sign of F determines whether or not a peg will fit in a hole.

$$E(F) = E(X) - E(Y) = 1.65 - 1.60 = 0.05$$

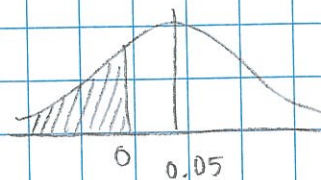
$$\text{Var}(F) = \text{Var}(X) + \text{Var}(Y) = 0.0004 + 0.0001 = 0.0005$$

$$F \sim N(0.05, 0.0005)$$

If for any combination of peg and hole the value of F is negative, then the peg will not fit into the hole.

hole < peg

$$\begin{aligned} P(F < 0) &= P\left(Z < \frac{0 - 0.05}{\sqrt{0.0005}}\right) \\ &= P(Z < -2.236) \\ &= 0.0127. \end{aligned}$$



COMBINING SEVERAL OBSERVATIONS OF THE SAME VARIABLE (ie $x+x+x+\dots+x$)

If x is a random variable, then the total of several (n) observations of the variable has mean and variance given by

$$E(x+x+x+\dots+x) = E(x) + E(x) + \dots + E(x) \\ = n \cdot E(x)$$

$$\text{Var}(x+x+x+\dots+x) = \text{Var}(x) + \text{Var}(x) + \text{Var}(x) + \dots + \text{Var}(x) \\ = n \cdot \text{Var}(x) \quad \neq \text{Var}(2x) = 2^2 \text{Var}(x)$$

Ex 1.

The mass, x , of a suitcase at an airport is modelled as being normally distributed, with mean 15 kg and standard deviation 3 kg. Find the probability that a random sample of ten suitcases weighs more than 154 kg.

Let the random variable T be the total weight of 10 suitcases

$$T = x + x + x + \dots + x$$

$$E(T) = E(x) + E(x) + \dots + E(x) = 15 + 15 + 15 + \dots + 15 = 150$$

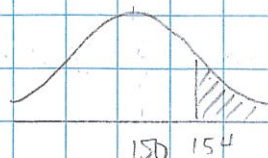
15×10
 $\uparrow$
 $3^2 \times 10$

$$\text{Var}(T) = \text{Var}(x) + \text{Var}(x) + \dots + \text{Var}(x) = 9 + 9 + 9 + \dots + 9 = 90$$

NOTE: $\text{Var}(x) + \text{Var}(x) + \dots + \text{Var}(x)$ is NOT the same as $\text{Var}(10x)$

So, the new variable T is $T \sim N(150, 90)$

$$P(T > 154) = P\left(Z > \frac{154 - 150}{\sqrt{90}}\right) = P(Z > 0.422) \\ = 0.337 \quad (3\text{sf})$$



Ex 2

The volume of liquid in cans of cola is normally distributed with mean 330 millilitres and standard deviation 5.2 millilitres. The volume of liquid in bottles of tonic water is normally distributed with mean 500 millilitres and standard deviation 7.1 millilitres.

(i) Find the prob. that 3 ^{randomly chosen} cans of cola contain less liquid than 2 randomly chosen bottles of tonic water.

Let C = vol in a random can of cola $C \sim N(330, 5.2^2)$

Let T = vol in a random bottle of Tonic $T \sim N(500, 7.1^2)$

We require $P(C+C+C < T+T)$

This is the same as $P(C+C+C-T-T < 0)$

If we let $D = C+C+C-T-T$, we require $P(D < 0)$

$$\begin{aligned}\text{Then } E(D) &= E(C+C+C-T-T) \\ &= E(C) + E(C) + E(C) - E(T) - E(T) \\ &= 330 + 330 + 330 - 500 - 500 \\ &= -10\end{aligned}$$

$$\begin{aligned}\text{And } \text{Var}(D) &= \text{Var}(C+C+C-T-T) \\ &= \text{Var}(C) + \text{Var}(C) + \text{Var}(C) + \text{Var}(T) + \text{Var}(T) \\ &= 5.2^2 + 5.2^2 + 5.2^2 + 7.1^2 + 7.1^2 \\ &= 181.94\end{aligned}$$

$$\begin{aligned}\text{So, } P(D < 0) &= P\left(Z < \frac{0 - (-10)}{\sqrt{181.94}}\right) \\ &= P(Z < 0.741) = 0.771.\end{aligned}$$

